

WEEK 7



Pyramid Variations

Hollow. Stepped. Your own.

피라미드 변형 — 속 빈·계단식·나만의 변형 · 영어코딩

BIG PICTURE

One pyramid, many remixes



Change a small part of the code → a brand new shape.

코드 한 부분만 바뀌도 새로운 모양

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Hollow = edges only

Place a block only when it's on the **edge** of the square.

Edge means: first row, last row, first column, or last column.

가장자리에만 블록을 놓으면 속이 빅니다

CODE

The edge condition

```
if row == 0 or row == size-1 or \
    col == 0 or col == size-1:
    agent.place(DOWN)
agent.move(FORWARD, 1)
```

On the edge → place. In the middle → skip.

WATCH IT!

Place only on the edge

edge cell

middle cell

row/col on edge?

✓ true → agent.place(DOWN)

✗ false → skip – leave hollow

▸ result

CODE

Stepped — bigger steps

```
size = 9
while size > 0:
    build_layer(size)
    agent.move(UP, 1)
    size = size - 4    # 9 → 5 → 1: chunky steps
```

Change the step number → change the look. `-4` = a stepped pyramid.

PREDICT 🤖

How many layers?

Predict first — then click reveal

```
size = 9
while size > 0:
    build_layer(size)
    agent.move(UP, 1)
    size = size - 4
```

Reveal answer 🔍 reveal

VOCAB

New words today

hollow empty inside — only the edges filled

edge the outer row or column of a square

if do something only when a test is true

stepped big jumps in size between layers

속이 빈

가장자리

만약 ~라면

계단식



What's wrong here?

buggy

```
# goal: a HOLLOW layer (edges only)  
for row in range(size):  
    for col in range(size):  
        agent.place(DOWN)  
        agent.move(FORWARD, 1)  
        agent.turn(RIGHT)
```

Think first — then click next.



DEBUG

The fix

Bug: It places a block on *every* cell, so the layer is solid, not hollow. Add an `if` so it only places on the edge.

fixed

```
if row==0 or row==size-1 or col==0 or col==size-1:  
    agent.place(DOWN)  
agent.move(FORWARD, 1)
```

YOUR TURN!

Invent a variation

- ✓ Pick one: hollow, stepped, gap, or your own
- ✓ Give it a name + a one-sentence description
- ✓ Change the part of the code that makes it different
- ✓ Build it in your homework world
- ✓ Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

What makes a pyramid layer **hollow**?

Placing blocks only on the edge cells.

Building it upside down.

Moving up twice each layer.

Using a smaller size.



QUIZ

QUESTION 2 OF 3

What does the `if` condition do in a hollow layer?

Places a block only when the cell is on the edge.

Moves the agent up.

Turns the agent right.

Counts the layers.



QUIZ

QUESTION 3 OF 3

size = 9, step size - 4 — how many layers?

5 layers.

9 layers.

3 layers (9, 5, 1).

2 layers.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Variations remixed!

Next: the final project — a giant diamond.

다음 주: 최종 프로젝트 — 거대한 다이아몬드